MIL-P-48764A (AR)
20 June 1979
SUPERSEDING
MIL-P-48764 (PA)
29 May 1975

MILITARY SPECIFICATION

PROJECTILE, 155MM, HE, ROCKET ASSISTED, M549 & M549A1, AFT (AHH) PROPELLANT GRAIN FOR

This specification is approved for use by the US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains requirements not covered by the drawings and provides quality assurance provisions for the fabrication of the Aft (AHH) Propellant Grain for the M549 and M549A1 Projectile.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-T-60	-	Tape, Packing, Water Proof
PPP-T-76	-	Tape, Pressure Sensitive Adhesive Paper, Water Resistant (For Carton Sealing)
PPP-F-320	-	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes

MILITARY

MIL-B-117	-	Bag, Sleeve and Tubing - Interior Packaging
MIL-C-13609	-	Casting Solvents for OIO, OGK, and AHH Propellants
MIL-C-45662	-	Calibration System Requirements
MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specification For
MIL-P-60432	-	Propellant Powder, Base Grain, AHH

FSC: 1320

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-286 - Propellants, Solid Sampling, Examination and Testing
MIL-STD-453 - Inspection, Radiographic
MIL-STD-652 - Propellants, Solid, for Cannons, Requirements and Packing

DRAWINGS (see 6.7)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

9298110 - Propellant, Grain Assembly, Aft

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with applicable drawings and specifications.

3.1.1 Casting powder. The casting powder shall comply with MIL-P-60432 (see 6.4). Except for the requirements for total volatiles and residual solvent this specification shall be advisory for a contractor that produces the AFT RAP grain.

3.1.2 Casting solvent. The casting solvent shall comply with MIL-C-13609, Type III. This requirement shall be advisory for a Contractor that produces the aft grain except that the nitroglycerin content shall be maintained at ± .10% for each propellant grain lot.

3.2 Propellant grain. The inhibited propellant grain shall comply with the requirements specified on Drawing 9298110 and with all requirements specified in applicable specifications.

3.3 Composition. The propellant grain composition shall be as specified in Table I when determined as specified in 4.5.1.

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TABLE I

Ingredient	Percentage
Nitrocellulose	53.9 MINIMUM
Nitroglycerin	31.6 NOMINAL
Triacetin	8.8 NOMINAL
2-Nitrodiphenylamine	0.90 MINIMUM
Lead Salicylate	1.2 NOMINAL
Lead 2-Ethylhexoate	2.0 NOMINAL

3.4 Heat of explosion. The heat of explosion shall be no less than 940 cal/gm and no more than 970 cal/gm when determined as specified in 4.4.3.1.4.

3.5 Strand burning rate

3.5.1 Burning rate. The propellant shall meet the burning rate specified in Table II when determined as specified in 4.4.3.2.

TABLE II

Temperature °F	Pressure P.S.I.	Burning Rate Inches/Seconds
-40	600	0.359 Minimum
+140	1600	0.581 Maximum

3.6 Cracks, foreign material and porosity

3.6.1 Radiographic examination:

a. Cracks - There shall be no cracks as shown on the radiograph as dark lines in excess of .06 inches in length.

b. Foreign material - There shall be no foreign material as shown by bright spots on the radiograph in excess of 0.06 inches in any shape or dimension.

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c. Porosity - There shall be no porous areas as shown visually by low density or rounded shadows on the radiograph in excess of .03 square inches individually or a maximum combined area of .06 square inches.

3.6.2 Visual examination:

a. Cracks - There shall be no cracks in excess of .06 inches in length.

b. Foreign material - There shall be no foreign material in excess of 0.06 inches in any shape or dimension.

c. Porosity - There shall be no porous areas in excess of .03 square inches individually or a maximum combined area of .06 square inches.

3.7 Inhibitor separation excluding inhibitor cone. The propellant grain shall contain no individual separation between the inhibitor and the grain or between the inhibitor layers (delamination) which is greater than 0.01 square inch. The total combined area of propellant inhibitor separations and inhibitor delaminations shall be no greater than 0.02 square inch. There shall be no propellant inhibitor separation within one inch of any split surface.

3.7.1 Inhibitor cone. No separation shall be allowable at the cut edges of the inhibitor cone.

3.8 Taliani test. The rate of evolution of gas from samples subjected to the test specified in 4.5.2 shall be such that the characteristics of a curve showing rate of evolution (millimeters of mercury versus time) meets the following:

a. Slope of the curve at 100mm of mercury shall be less than 1.00.
b. Slope of the curve at 100 minutes shall be less than 0.90.
c. Time to 100mm of mercury shall be more than 100 minutes.

3.9 120°F heat test. The propellant shall not cause complete fading of methyl violet paper to a salmon pink color in less than 60 minutes when tested as specified in 4.5.3.

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3.10 Physical properties. Compression load and deformation.

3.10.1 Plus 140°F. The average of three (3) maximum compression readings from each casting shall be no less than 870 lbs., and the average of three (3) deformation readings from each casting shall be no less than 53 percent when tested as specified in 4.5.4.

3.10.2 Minus 40°F. The average of three (3) compression readings from each casting shall be no less than 12,580 pounds and the average of three (3) deformation readings from each casting shall be no less than 5.25 percent when tested as specified in 4.5.4.

3.11 Workmanship

3.11.1 Propellant grain assembly. The propellant grain assembly shall be free of dirt, grease, chips, gouges, solvent drippings, and other defects which would make the item unsuitable for its intended use.

3.11.2 Inhibitors. The inhibitors (cone and circumferential inhibitor) shall be free from dirt, grease, chips, gouges, cracks, and other defects which would make the item unsuitable for its intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First article inspection

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

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Part Description	Drawing	Quantity
Inhibitor, Circumferential, Prior to Grain Casting	9298111	10
Inhibitor Cone	9298112	10
Propellant Grain Aft, Before Adding Inhibitor Cone	9298111	10
Propellant Grain Assembly Aft, Before Splitting	9298110	10
Propellant Grain Assembly Aft, After Splitting	9298110	10
Propellant - Special Lot Test Samples	See Table III	See Table III
Propellant Grains, Aft (For Ballistic Testing, See 4.3.2.2.1 for Info Only)	9298110	60

4.3.2 Inspections to be performed. See MIL-A-48078 and Table III specified herein.

4.3.2.1 Ballistic tests

4.3.2.1.1 Submission. Sixty complete propellant grains packed in the form that will be used for shipping of the regular production quantity shall be submitted to a Government Installation as designated by the Contracting Officer for loading and ballistic testing. Unless otherwise directed by the Contracting Officer, the propellant grains shall not be submitted prior to the acceptance of the sample submitted to the Government for examination (see 4.3.1).

4.3.2.1.2 Tests to be performed. Proving Ground tests shall be performed for information only using the procedures of 4.5.8.

4.3.3 Rejection. See MIL-A-48078.

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TABLE III- FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48764A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
	Aft (AHH) Propellant Grain				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Inhibitor, Circumferential, Prior to Grain Casting (Dwg. 9298111) Examination for Defects	10	-	3.2	4.4.2.1
	Inhibitor Cone (Dwg. 9298112) Examination for Defects	10	-	3.2	4.4.2.2
	Propellant Grain Aft, Before Adding Inhibitor Cone (Dwg. 9298111) Examination for Defects	10	-	3.2	4.4.2.3
	Propellant Grain Assembly Aft, Before Splitting (Dwg. 9298110) Examination for Defects	10	-	3.2	4.4.2.4

NOTES:

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TABLE III- FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48764A (AR)

PARAGRAPH	TITLE	SHEET 2 OF 2			DRAWING NUMBER
	CONTINUATION SHEET				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Propellant Grain Assembly Aft, After Splitting (Dwg. 9298110)				
	Examination for Defects	10	-	3.2	4.4.2.5
	Radiographic Inspection	10	-	3.6.1	4.5.6
	Special Lot Tests				
	Chemical Composition	3	-	3.3	4.5.1
	Taliani Test	3	-	3.8	4.5.2
	120° Heat Test	3	-	3.9	4.5.3
	Heat of Explosion	3	-	3.4	4.5.7
	Physical Properties - Plus 140°F	3	-	3.10.1	4.5.4
	Physical Properties - Minus 40°F	3	-	3.10.2	4.5.4
	Strand Burning Rate	24 Strands	-	3.5	4.5.5

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4.4 Quality conformance inspection

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of Propellant Grains shall contain:

a. Not more than one lot of casting powder. The lot interfix number shall be changed when a new lot of casting powder is introduced into the rocket grain casting processes.
b. No less than 1 weeks production of rocket grains or more than 1 months production of rocket grains.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48764A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Inhibitor, Circumferential, Prior to Grain Casting		9298111
			NEXT HIGHER ASSEMBLY 9298110

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Thickness of inhibitor wall		0.65%	3.2	Gage
202	Evidence of poor workmanship		1.0%	3.11.2	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48764A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.2	Inhibitor, Cone		9298112
			NEXT HIGHER ASSEMBLY
			9298110

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Overall length (Max.)		0.40%	3.2	Gage
102	Angle of cone		0.40%	3.2	Gage
103	Thickness		0.40%	3.2	Gage
104	Evidence of poor workmanship		0.40%	3.11.2	Visual
Minor	None defined				

NOTES:
See 4.3.3.7.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48764A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.3	Propellant Grain Aft, Before Adding Inhibitor Cone		9298111
			NEXT HIGHER ASSEMBLY 9298110

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Location of small basic diameter		.40%	3.2	Gage
102	Location of large basic diameter		.40%	3.2	Gage
103	Total length, max.		.40%	3.2	Gage
104	True position of inside diameter to outside diameter		.40%	3.2	Gage
105	Outside diameter of inhibitor				
Minor					
201	Inside diameter		.65%	3.2	Gage
202	Diameter of inside diameter chamfer		.65%	3.2	Gage
203	Evidence of poor workmanship		1.0%	3.11	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48764A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Propellant Grain Assembly Aft, Before Splitting	1 OF 1	9298110
			NEXT HIGHER ASSEMBLY 9235999

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Poor joint or improper blending of top and bottom cone junction with bottom and side inhibitor		0.40%	3.2	Visual
102	Improper radius		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.0%	3.11	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48764A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.5	Propellant Grain Assembly Aft, After Splitting		9298110
			NEXT HIGHER ASSEMBLY 9235999

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Excessive cracks, foreign material, or porosity		100%	3.6.2	Visual
102	Exposed propellant at intersections of inhibitor surfaces		100%	3.2	Visual
103	Surface condition of saw cut		100%	3.2	Visual
104	Inhibitor separation		0.40%	3.7	Visual
105	Finished weight of grain		0.40%	3.2	Balance
106	Spacer missing		0.40%	3.2	Visual
107	Cone separation		0.40%	3.7.1	Visual
Minor					
201	Evidence of poor workmanship		1.0%	3.11	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.6	Packing, Aft Grain		NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Drum split or damaged		0.40%	5.1	Visual
102	Tape missing or improperly applied		0.40%	5.1	Visual
Minor					
201	Marking incomplete or incorrect		0.65%	5.1	Visual
202	Evidence of poor workmanship		1.0%	3.11	Visual

NOTES:
See 5.1 and 5.2.

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4.4.3 Testing

4.4.3.1 Lot tests

4.4.3.1.1 Chemical composition - Major defect. Three samples in accordance with 4.4.3.1.5 shall be randomly selected from the lot and tested in accordance with 4.5.1. If any sample fails to meet the requirements of 3.3, the lot shall be rejected.

4.4.3.1.2 Taliani test - Major defect. Three samples in accordance with 4.4.3.1.5 shall be randomly selected from the lot and tested in accordance with 4.5.2. If any sample fails to meet the requirements of 3.8, the lot shall be rejected.

4.4.3.1.3 120° heat test - Major defect. Three samples in accordance with 4.4.3.1.5 shall be randomly selected from the lot and tested in accordance with 4.5.3. If any sample fails to meet the requirements of 3.9, the lot shall be rejected.

4.4.3.1.4 Heat of explosion - Major defect. Three samples in accordance with 4.4.3.1.5 shall be randomly selected from the lot and tested in accordance with 4.5.7.

4.4.3.1.4.1 Rejection. If any one of the three samples fail to meet the requirements of 3.4, one sample shall be submitted from each of the six additional different propellant sublots. If any of the six samples fail to meet the requirements of 3.4, the lot shall be rejected.

4.4.3.1.5 Sampling procedures. The samples for the tests of 4.4.3.1.1, 4.4.3.1.2, 4.4.3.1.3 and 4.4.3.1.4 shall be drawn as three samples and then each sample subdivided so as to produce specimens for each of the four tests. If possible, no two samples will be taken from the same casting solvent sublot (mix.).

4.4.3.2 Strand burning rate - Major defect. Twenty-four strands shall be formed from each of three different propellant casting sublots and tested in accordance with 4.5.5. Three strands shall be used to determine the average burning rate at each temperature pressure combination as specified in paragraph 3.5.1.

4.4.3.2.1 Rejection. If any one of the average burning rates from three castings fail to meet the requirements of 3.5.1, six additional propellant castings from different sublots shall be

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selected. A total of eighteen shall be used to determine the average burning rate at each mandatory temperature pressure combination specified in 3.5.1. If any one of the average burning rates from any casting fails to meet the requirements of 3.5.1, the lot shall be rejected.

4.4.3.3 Radiographic inspection

4.4.3.3.1 Lot inspection - Major defect. One sample of a completed grain shall be selected from each sublot produced shall b radiographed. A sublot shall be defined as a group of castings, cast simultaneously from the same casting liquid batch. In the event that the sample grain fails to meet the requirements of 3.6.1 the entire sublot shall be x-rayed and any additional grains failin to meet the requirements of 3.6.1 shall be removed from the lot. This test shall be conducted in accordance with 4.5.6.

4.4.3.4 Physical properties

4.4.3.4.1 Plus 140°F (see 3.10.1) - Major defect. Three samples will be randomly selected from the lot. If possible, no tw samples will be taken from the same casting solvent sublot. These samples shall be tested for maximum compression load and deformatio as specified in 4.5.4. The lot shall be rejected if the average of the three specimens from any casting fails to meet the requirements of 3.10.1.

4.4.3.4.2 Minus 40°F (see 3.10.2) - Major defect. Three samples will be randomly selected for the lot. If possible, no two samples will be taken from the same casting solvent sublot. These samples shall be tested for compression load and deformation as specified in 4.5.4. The lot shall be rejected if the average of t three specimens from any casting fails to meet the requirements of 3.10.2.

4.4.3.5 Width of cut - Major defect. The width of cut taken splitting the grain shall not exceed the drawing requirement. For verification, the width of cut shall be inspected at the beginning of each 8 hours of production.

4.4.3.6 Inhibitor, Cone - Major defect - (Drawing 9298112). T following sampling sequence shall be used for each cavity of each mold or each set of dies as appropriate. The occurrence of any defect on any sample piece shall be cause for taking corrective action and for rejecting all pieces produced since the last acceptable sample.

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4.4.3.6.1 The first twenty (20) consecutive pieces from each set of dies or mold cavity shall be inspected for dimensional characteristics as shown on Dwg. 9298112 and 4.4.2.2 as well as for workmanship.

4.4.3.6.2 Five (5) pieces from each set of dies or mold cavity at intervals of 1,000 pieces from each set of dies or mold cavity shall be inspected for dimensional characteristics as specified on Dwg. 9298112 and 4.4.2.2 as well as visual examination for stamping or molding defects and poor workmanship.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the 'Paragraph Reference/Inspection Method' column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection

4.5.1 Composition

4.5.1.1 Nitrocellulose. The nitrocellulose content shall be determined in accordance with Standard MIL-STD-286, Method 209.2 or Method 209.6.

4.5.1.2 2-Nitrodiphenylamine. The 2-nitrodiphenylamine content shall be determined in accordance with Standard MIL-STD-286, Method 218.4.

4.5.1.3 Nitroglycerin. The nitroglycerin content shall be determined in accordance with Standard MIL-STD-286, Method 208.1.

4.5.1.4 Triacetin. The triacetin content shall be determined in accordance with Standard MIL-STD-286, Method 222.1.

4.5.1.5 Lead salicylate and lead 2 - Ethylhexoate. Total lead content shall be determined in accordance with Standard MIL-STD-286, Method T316.1; lead salicylate content shall be determined in accordance with MIL-STD-286, Method 225.1; lead 2-ethylhexoate shall be determined by difference.

4.5.2 Taliani test. This test shall be performed in accordance with Standard MIL-STD-286, Method 406.1.

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4.5.3 120°C heat test. This test shall be performed in accordance with Standard MIL-STD-286, Method 404.1.

4.5.4 Physical properties. Each specimen and specimen support shall conform to the requirements of Figure 1. The compression testing shall be conducted in a test fixture equipped with a temperature conditioning box as an integral part of the fixture. The specimen shall be inserted between the fixture posts with steel cylinders the same diameter as the specimen and placed above and below the specimen forming one continuous concentric cylinder. The specimen shall be conditioned at the required temperature plus or minus 5°F for three hours minimum and shall be tested at a crosshead speed of 2 inches per minute.

4.5.5 Strand burning rate - (4.4.3.2). The strand burning rate shall be determined in accordance with Method T803.1 of Standard MIL-STD-286 for test in nitrogen (-40°F) and a Government approved Contractor method for testing in oil (SAE #10, for 70°F and 140°F tests). The strands tested in oil shall be uninhibited.

4.5.6 Radiographic inspection. The split finished machined propellant grain shall be radiographed with the beam parallel to the longitudinal axis of the grain. Observations shall be made for cracks, voids, fissures and foreign material in accordance with 3.6.1. Radiographic inspection procedures shall be in accordance with MIL-STD-453. A penetrameter or sample test block showing the types of defects for rejection shall be designed to assure a 2 percent minimum sensitivity.

4.5.7 Heat of explosion - (See 4.4.3.1.4). This test shall be performed in accordance with Standard MIL-STD-286, Method 802.1.

4.5.8 Ballistic acceptance test - (First Article). A ballistic acceptance test using the standard testing procedures and acceptance criteria of MIL-P-50578 for Complete Round Loading, Assembly and Packing shall be performed for information only. All test data and raw magnetic radar tapes shall be sent to Commander, ARRADCOM, Dover, N.J. 07801, ATTN: DRDAR-QAR-Q.

5. PACKAGING

5.1 Packing - Level B. Twelve (12) aft grains shall be packed in a fiber drum meeting the requirements of DOT specifications 21C of the Code of Federal Regulations 49 CFR Parts 100 to 199. The

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drum wall shall be lined with a kraft backed commercial grade single faced corrugated fiber board using as many layers as required for a tight pack. The drum size shall be approximately 11 5/8 by 11 5/8 by 9 inches high. A barrier bag meeting requirements of MIL-B-117, Type I, Class E, Style 1 shall be placed inside the lined drum and 12 propellant grains shall be packed in 3 layers of 4 per layer. The barrier bag size shall be approximately 24 by 22 1/4 inches. Each grain shall be inserted into a spiral or convolute wound chipboard tube having the following dimensions: Inside Diameter, 4.650 + .015; Outside Diameter, 4.975 - .030; Length 2.860 - .020 inches. The layers of the chipboard tube shall be adhered with dextrin or polyvinyl adhesive and the inside wall surface shall be smooth. Each grain shall be placed in the fiber drum with the marked end up. A separator shall be placed between layers and on top of final layer. The separator shall meet the requirements of PPP-F-320, Type CF, Class Domestic, Grade 200, B or C Flute material or equivalent. The bag shall be folded and sealed with tape meeting requirements of PPP-T-60 or equivalent. Filler material constructed of the same material as the separators shall be placed on top of the contents as required to provide a tight pack. The fiber drum cover shall be sealed with 2 inch minimum wide tape meeting the requirements of PPP-T-76 or equivalent. All packing material inside the barrier bag shall not exceed 7 percent maximum moisture content.

5.2 Marking. The fiber drum shall be marked as follows:

DOT marking on top - "PROPELLANT EXPLOSIVES (SOLID) CLASS B"

On Sidewall - 12 GRAIN, PROPELLANTS, AFT f/RAP 155MM, M549
DATE PACKED (MO/YR)
LOT NO (INSERT NO)
WT (INSERT WT) CU (INSERT VOLUME)

Marking shall be with black waterproof ink and lettering shall be 1/4 inch minimum height.

6. NOTES

6.1 Intended use. The components covered by this specification are intended for use on the 155MM HE Rocket Assisted Projectile, M549 and M549A1.

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6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment for design approvals - See MIL-A-48078. Submit designs as required to Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-I, Dover, New Jersey 07801.

6.4 Base grain powder test. In the event AHH base grain powder has been in storage for one year or longer, the powder shall be requalified in accordance with Specification MIL-P-60432.

6.5 Submission of test data. In addition to the normal distribution of records, one copy of all static and compression test data will be forwarded to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-Q, Dover, New Jersey 07801.

6.6 Surveillance sample. Samples shall be forwarded to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-LCE-M, Dover, New Jersey 07801 for the propellant surveillance test program in accordance with MIL-STD-652. This sample shall contain propellant pieces not exceeding one inch in diameter which may include machine turnings, broken tensile specimens, or pieces of grain segments, but no inhibitor materials are to be included.

6.7 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

6.9 Previous issues. Previous versions of MIL-P-48764, having been issued as classified documents strictly for the use of the preparing activity, were not submitted to the Department of Defense Single Stock Point (DODSSP) and, therefore, are not available from DODSSP.

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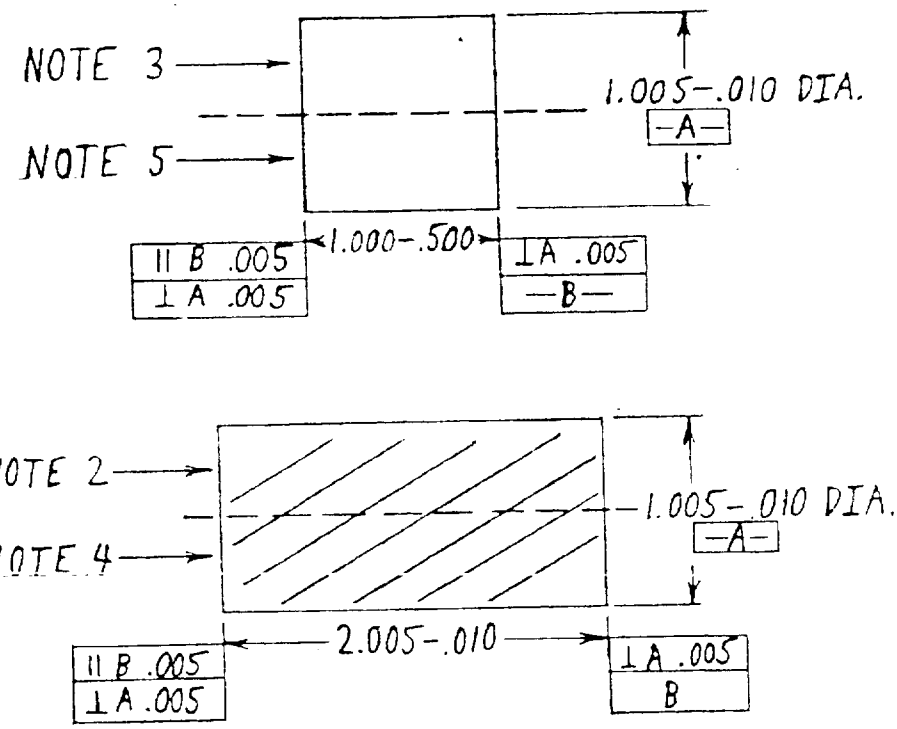
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APPLICABLE SPECIFICATIONS-MIL-P-48764A(AR
-MIL-P-50573C(AR)

FIGURE 1



NOTES:

1. SPECIFICATION MIL-A-2550 APPLIES.
2. SPECIMEN MATERIAL-SOLID PROPELLANT.
3. TWO STEEL SPECIMEN SUPPORT REQUIRED & SHALL BE PLACED ABOVE & BELOW SPECIMEN PRIOR TO TEST.
4. SURFACE FINISH-125/ALL OVER.
5. SURFACE FINISH-125/ON PROPELLANT SURFACE.

TITLE:
RAP GRAIN SPECIMEN & SPECIMEN SUPPORTS FOR COMPRESSION TEST

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-P-48764A Projectile, 155MM, HE, Rkt Assisted, M549 & M549A1

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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